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RESEARCH ARTICLE

Extended ExpTODIM Framework Based on Improved GRA Approach for Comprehensive Quality Evaluation of Physical Education Institutions Under Hesitant Triangular Fuzzy Sets

RUI LU, GUOXIAN GAO, AND YUHONG XIANG
School of Physical Education, Guizhou University of Finance and Economics, Guiyang, Guizhou 550000, China
Corresponding author: Rui Lu (tigertiger67890@163.com)

ABSTRACT Under the new educational philosophy, the comprehensive quality evaluation of students in sports colleges is increasingly receiving attention from educators and society. Because comprehensive quality not only covers professional knowledge and skills, but also includes moral qualities, innovative spirit, teamwork ability, etc., which are all urgent needs for talents in sports colleges and universities in society. However, the comprehensive quality evaluation system for students in sports colleges and universities seems to be unable to fully meet this demand, and there are some problems that cannot be ignored, such as excessive emphasis on grades, fixed evaluation systems, lack of student participation in the evaluation process, and fuzzy evaluation standards. The existence of these problems not only affects the fairness and effectiveness of evaluation, but also hinders the comprehensive development of students and the process of educational reform. Therefore, how to optimize the comprehensive quality evaluation system of students in sports colleges and universities to better adapt to the needs of social development has become an important topic in current sports education research. The comprehensive quality evaluation of physical education institutions is multi-attribute decision-making (MADM). Exponential TODIM (ExpTODIM) and improved grey relational analysis (IGRA) approaches were used to address MADM issues in this context. Hesitant Triangular Fuzzy Sets (HTFSs) have been employed as a tool for characterizing the uncertain information inherent in the comprehensive quality evaluation of physical education institutions. In this paper, we develop a Hesitant Triangular Fuzzy Number Exponential TODIM-IGRA (HTFN-ExpTODIM-IGRA) approach to illustrate the MADM problem under the framework of HTFSs. Finally, numerical study for the comprehensive quality evaluation of physical education institutions, along with comparative analysis, is provided to validate the HTFN-ExpTODIM-IGRA approach.

INDEX TERMS Multiple-attribute decision-making (MADM), HTFSs, ExpTODIM approach, improved GRA approach, capability evaluation.

I. INTRODUCTION

The comprehensive quality of students is a dynamic process, and evaluation at the end of the semester alone will lose the concrete manifestation of the development achievements of students' comprehensive quality [1], [2]. It is necessary to organically combine summative, procedural, developmental, and formative evaluations to achieve the goal of sustainable development for students [3], [4]. The internship course for students in sports colleges is an important means for students to enhance their social and professional abilities. Therefore, collaborative evaluation of students' comprehensive quality changes during both on campus learning and off campus internships is an important prerequisite for ensuring the timeliness and accuracy of evaluation results [5], [6]. Meanwhile, the changing situation of students is complex

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and difficult to judge. Therefore, the diagnostic evaluation, formative evaluation, and summative evaluation of students' comprehensive qualities should be dynamically updated and given feedback during the learning process [7], [8]. Ensuring dynamic feedback on the evaluation results of students' comprehensive qualities in both on and off campus teaching processes is the primary task of guiding students' self-growth and teachers' teaching reform. Adhering to the dynamic diagnosis and synchronous correction of sports colleges in the process of cultivating students' comprehensive qualities is one of the main goals of evaluating students' comprehensive qualities [9], [10]. The top-level planning and design of the school, the curriculum system design of the college, and the content design of the teacher's curriculum should all be personalized, refined, and accurately dynamically adjusted based on the developmental process of students' comprehensive qualities, maximizing the use of data on the development and changes of students' comprehensive qualities to support the teaching quality update mechanism. By continuously updating the training mode, we aim to tap into students' own potential and encourage them to make up for their own shortcomings [11], [12]. At the same time, universities should introduce corresponding detailed rules for the evaluation and cultivation of students' comprehensive qualities, and promote the evaluation and cultivation of comprehensive qualities in sports colleges from the perspective of university management [13], [14]. However, from the published comprehensive quality management measures for sports colleges and universities, it can be seen that there are no sports colleges and universities in some region that have implemented this work, which is a gap in research within the region [15], [16]. At the same time, the cultivation and evaluation of students' comprehensive qualities should not only be emphasized in compulsory and secondary education. Sports colleges shoulder the important task of exporting skilled talents in China, and the cultivation of comprehensive qualities, as an important prerequisite for the personal development of skilled talents, should be given more attention by the government [17], [18]. The government should pay attention to the promotion of comprehensive quality cultivation and evaluation policies for students in higher physical education institutions in the context of the new vocational education law [19], [20]. Nowadays, many sports colleges and universities habitually use paper tools or enter grades in the final assessment process for comprehensive quality evaluation [21], [22]. The poor timeliness of information collection leads to the lack of guiding significance in the evaluation results. At the same time, in corporate practice and family observation, evaluators find it difficult to rate in real expression scenarios [23]. In the context of the Big data era, it is a model recommended by scholars to use the information system to improve the efficiency of comprehensive quality evaluation and dynamic feedback evaluation results of students in sports colleges [24], [25]. Select appropriate evaluation methods from the constructed indicator system and open different evaluation terminals for different evaluators. Personnel with different levels of authority can view the comprehensive quality scores and dynamic changes of students within their responsible blocks, and can form corresponding ''comprehensive quality archives'' for individual students. This is beneficial for counselors and teachers to guide students' development in a targeted manner, and also helps the leadership adjust top-level design based on changes in students' comprehensive quality [26], [27].

Decision analysis widely exists in politics, economy, technological development, and daily life. It often calculates, analyzes, and judges the optimal factors that affect the achievement of goals based on objective possibilities and certain information and experience, in order to make predictions for future actions [28], [29], [30], [31]. Generally speaking, a complete decision-making process includes identifying problems, determining research objectives, determining evaluation criteria and their weights, scheme formulation, scheme selection, and scheme implementation [32], [33], [34], [35], [36]. Decision analysis is a valuable tool that helps people improve the quality of their decision-making while reducing the time and cost associated with the process. However, as human society, economy, and other activities continue to evolve, the decision-making landscape has become increasingly complex. The number of unknown influencing factors in the decision-making process has rapidly increased, and the depth and breadth of knowledge required continues to expand [37], [38], [39], [40], [41]. As a result, the initial precise mathematical models are no longer sufficient to adequately characterize the uncertainty inherent in practical decision-making problems. Therefore, proposing more effective ways of expressing information and developing more robust decision-making methods has become an inevitable requirement for the ongoing advancement of modern decision-making science [42], [43], [44]. ADM, as important decision branch of decision science, is commonly used to handle group decision-making (GDM) problems in discrete decision spaces. MADM could deal with both qualitative and quantitative evaluation data and fuzzy uncertainty information, so it has wide applications in Military engineering, economic science, logistics supply and other fields [45], [46], [47], [48], [49]. In the framework of MADM problems, decision makers (DMs) often take multiple attributes as the evaluation criteria, comprehensively investigate the evaluation object to achieve optimal selection. Scholars have proposed MADM theory based on social axiom of choice and other theories, and then introduced fuzzy mathematics into MADM and GDM theory [50], [51], [52], [53], [54]. At present, DMs have applied fuzzy control, pattern recognition, medical diagnosis and other aspects have been further developed [55], [56], [57], [58].

The comprehensive quality evaluation of physical education institutions is a MADM. Currently, ExpTODIM [43], [59] and improved GRA approaches [60], [61], [62], [63], [64] have been used to address MADM issues in this context. HTFSs [65] have been employed as framework for characterizing the uncertain data inherent in the comprehensive

quality evaluation of physical education institutions. However, to date, ExpTODIM-IGRA approach with HTFS has not been administrated. Thus, it is important to investigate the HTFN-ExpTODIM-IGRA approach, which combines the benefits of the original ExpTODIM and IGRA approaches with the HTFS framework. The aim of this work is to develop an enhanced integrated model that incorporates the ExpTODIM and IGRA approaches within the HTFS information framework, in order to address MADM more effectively. Specifically, the HTFN-ExpTODIM-IGRA approach is illustrated for the MADM problem with HTFSs. Finally, numerical study for the comprehensive quality evaluation of physical education institutions is presented to validate the proposed approach. The research goals and motivations of this paper are summarized as follows: (1) The development of HTFN-ExpTODIM-IGRA approach for MADM problems with HTFS information; (2) The utilization of MEREC approach to administrate the attribute weight information; (3) The application of HTFN-ExpTODIM-IGRA approach to the comprehensive quality evaluation of physical education institutions, and a comparative analysis with existing approaches; (4)The demonstration of the effectiveness of the proposed HTFN-ExpTODIM-IGRA approach for comprehensive quality evaluation of physical education institutions through the comparative analysis. By integrating the ExpTODIM and IGRA methods within the HTFS framework, this research provides a robust and comprehensive decision-making tool for evaluating the quality of physical education institutions, accounting for the inherent uncertainties in the MADM process.

The research structure of this study is outlined: Section II introduces the HTFSs and its mathematical formulation. In Section III, we design the HTFN-ExpTODIM-IGRA approach under the HTFS framework. Section IV presents numerical study on the comprehensive quality evaluation of physical education institutions, along with comparative analysis of results administrated using the HTFN-ExpTODIM-IGRA approach. Finally, Section V offers some concluding remarks.

II. PRELIMINARIES

Wei et al. [65] constructed the HTFSs.

Definition 1 [65]: The HTFSs is constructed:

$$V = \{\langle \theta, u_V(\theta)\rangle \,|\theta \in \Theta\} \tag{1}$$

where $\mu_V(\theta) \subset [0,1]$ is triangular fuzzy numbers which are possible membership of $\theta \in \Theta$, $\forall\ \theta \in \Theta$. Then, $v\theta = (vl, vm, vr)$ is denoted HTFN.

Definition 2 [65]: Let $v\theta_1 = (vl_1, vm_1, vr_1)$ and $v\theta_2 = (vl_2, vm_2, vr_2)$ be HTFNs, the operations are constructed:

$$v\theta_1 \oplus v\theta_2 = \bigcup_{o\theta_1 \in v\theta_1, o\theta_2 \in v\theta_2} \left\{ \begin{pmatrix} ol_1 + ol_2 - ol_1 \times ol_2, \\ om_1 + om_2 - om_1 \times om_2, \\ or_1 + or_2 - or_1 \times or_2 \end{pmatrix} \right\} \tag{2}$$

$$v\theta_1 \otimes v\theta_2 = \bigcup_{o\theta_1 \in v\theta_1, o\theta_2 \in v\theta_2} \{(ol_1 \times ol_2, om_1 \times om_2, or_1 \times or_2)\} \tag{3}$$

$$\lambda \times v\theta_1 = \bigcup_{o\theta_1 \in v\theta_1} \left\{ \left(1-(1-ol_1)^{\lambda}, 1-(1-om_1)^{\lambda}, 1-(1-or_1)^{\lambda}\right)\right\},\ \lambda > 0 \tag{4}$$

$$(v\theta_1)^{\lambda} = \bigcup_{o\theta_1 \in v\theta_1} \left\{ \left((ol_1)^{\lambda}, (om_1)^{\lambda}, (or_1)^{\lambda}\right)\right\},\ \lambda > 0 \tag{5}$$

From Definition. 2, the operation laws are administrated.

$$\begin{aligned}
(1)\ v\theta_1 \oplus v\theta_2 &= v\theta_2 \oplus v\theta_1,\ v\theta_1 \otimes v\theta_2 \\
&= v\theta_2 \otimes v\theta_1,\ \left((v\theta_1)^{\lambda_1}\right)^{\lambda_2} \\
&= (v\theta_1)^{\lambda_1\lambda_2}; \\
(2)\ \lambda(v\theta_1 \oplus v\theta_2) &= \lambda v\theta_1 \oplus \lambda v\theta_2,\ (v\theta_1 \otimes v\theta_2)^{\lambda} \\
&= (v\theta_1)^{\lambda} \otimes (v\theta_2)^{\lambda}; \\
(3)\ \lambda_1 \times v\theta_1 \oplus \lambda_2 \times v\theta_1 &= (\lambda_1+\lambda_2)\, v\theta_1,\ (v\theta_1)^{\lambda_1} \otimes (v\theta_1)^{\lambda_2} \\
&= (v\theta_1)^{(\lambda_1+\lambda_2)}.
\end{aligned}$$

Definition 3 [65]: Let $v\theta_1 = (vl_1, vm_1, vr_1)$ and $v\theta_2 = (vl_2, vm_2, vr_2)$ be HTFNs, the score functions of $v\theta_1 = (vl_1, vm_1, vr_1)$ and $v\theta_2 = (vl_2, vm_2, vr_2)$ is constructed:

$$SF(v\theta_1) = \frac{1}{\#v\theta_1} \sum\nolimits_{o\theta_1 \in v\theta_1} \frac{ol_1 + 2om_1 + or_1}{4} \tag{6}$$

$$SF(v\theta_2) = \frac{1}{\#v\theta_2} \sum\nolimits_{o\theta_2 \in v\theta_2} \frac{ol_2 + 2om_2 + or_2}{4} \tag{7}$$

where $\#v\theta_1$ and $\#v\theta_2$ are numbers of $v\theta_1 = (vl_1, vm_1, vr_1)$ and $v\theta_2 = (vl_2, vm_2, vr_2)$.

For two HTFNs $v\theta_1 = (vl_1, vm_1, vr_1)$ and $v\theta_2 = (vl_2, vm_2, vr_2)$, then

$$\begin{aligned}
&(1)\ \textit{if}\ SF(v\theta_1) < SF(v\theta_2),\quad v\theta_1 < v\theta_2; \\
&(2)\ \textit{if}\ SF(v\theta_1) = SF(v\theta_2),\quad v\theta_1 = v\theta_2.
\end{aligned}$$

Definition 4 [66]: Let $v\theta_1 = (vl_1, vm_1, vr_1)$ and $v\theta_2 = (vl_2, vm_2, vr_2)$ be HTFNs, the Euclidean distance is constructed: (8), as shown at the bottom of the next page, where $ol_1(\sigma(k)), om_1(\sigma(k)), or_1(\sigma(k)), ol_2(\sigma(k)), om_2(\sigma(k))$ and $or_2(\sigma(k))$ are kth largest values in $v\theta_1 = (vl_1, vm_1, vr_1)$ and $v\theta_2 = (vl_2, vm_2, vr_2)$ and $\#v\theta = \#v\theta_1 = \#v\theta_2$.

The HTFWA and HTFWG approach is built [67].

Definition 5 [67]: Let $v\theta_j = (vl_j, vm_j, vr_j)$ $(j = 1, 2, \cdots, n)$ be HTFNs, the HTFWA approach is:

$$HTFWA_{v\omega}(v\theta_1, v\theta_2, \ldots, v\theta_n) = \bigoplus_{j=1}^{n} \left(v\omega_j \otimes v\theta_j\right)$$

$$= \bigcup_{o\theta_1 \in v\theta_1} \left\{ \begin{pmatrix} 1 - \prod_{j=1}^{n} \left(1 - ol_j\right)^{v\omega_j} \\ 1 - \prod_{j=1}^{n} \left(1 - om_j\right)^{v\omega_j}, \\ 1 - \prod_{j=1}^{n} \left(1 - or_j\right)^{v\omega_j} \end{pmatrix} \right\} \tag{9}$$

where $v\omega = (v\omega_1, v\omega_2, \ldots, v\omega_n)^T$ be weight of $v\theta_j$, $v\omega_j > 0$, $\sum_{j=1}^{n} v\omega_j = 1$.

Definition 6 [67]: Let $v\theta_j = \left(vl_j, vm_j, vr_j\right)$ $(j = 1, 2, \cdots, n)$ be HTFNs, the HTFWG approach is:

$$\begin{aligned} &HTFWG_{v\omega}(v\theta_1, v\theta_2, \ldots, v\theta_n) \\ &= \bigotimes_{j=1}^{n} \left(v\theta_j\right)^{v\omega_j} \\ &= \bigcup_{o\theta_j \in v\theta_j} \left\{ \left(\prod_{j=1}^{n} \left(ol_j\right)^{v\omega_j}, \prod_{j=1}^{n} \left(om_j\right)^{v\omega_j}, \prod_{j=1}^{n} \left(or_j\right)^{v\omega_j} \right) \right\} \end{aligned} \tag{10}$$

where $v\omega = (v\omega_1, v\omega_2, \ldots, v\omega_n)^T$ be weight of $v\theta_j$, $v\omega_j > 0$, $\sum_{j=1}^{n} v\omega_j = 1$.

III. HTFN-ExpTODIM-IGRA APPROACH FOR MADM WITH MEREC

A. HTFNMAGDM DESCRIPTION

The HTFN-ExpTODIM-IGRA approach is illustrated for MADM. Let $VA = \{VA_1, VA_2, \cdots, VA_m\}$ be alternatives, and $VG = \{VG_1, VG_2, \cdots, VG_n\}$ be attributes with weight $v\omega = (v\omega_1, v\omega_2, \ldots, v\omega_n)^T$, $v\omega_j > 0$, $\sum_{j=1}^{n} v\omega_j = 1$.

Then, HTFN-ExpTODIM-IGRA approach is built for MADM (See Figure 1).

Step 1: Build the HTFN-matrix $VR = \left[v\theta_{ij}\right]_{m\times n} = \left(vl_{ij}, vm_{ij}, vr_{ij}\right)_{m\times n}$:

$$VR = \left[v\theta_{ij}\right]_{m\times n} = \begin{array}{c} \\ VA_1 \\ VA_2 \\ \vdots \\ VA_m \end{array} \begin{array}{c} \begin{array}{cccc} VG_1 & VG_2 & \ldots & VG_n \end{array} \\ \begin{bmatrix} v\theta_{11} & v\theta_{12} & \ldots & v\theta_{1n} \\ v\theta_{21} & v\theta_{22} & \ldots & v\theta_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ v\theta_{m1} & v\theta_{m2} & \ldots & v\theta_{mn} \end{bmatrix} \end{array} \tag{11}$$

Step 2: Normalize the $VR = \left[v\theta_{ij}\right]_{m\times n} = \left(vl_{ij}, vm_{ij}, vr_{ij}\right)_{m\times n}$ into $NVR = \left[nv\theta_{ij}\right]_{m\times n} = \left(nvl_{ij}, nvm_{ij}, nvr_{ij}\right)_{m\times n}$.

For benefit attributes:

$$nv\theta_{ij} = \left(nvl_{ij}, nvm_{ij}, nvr_{ij}\right) = \left(vl_{ij}, vm_{ij}, vr_{ij}\right) \tag{12}$$

For cost attributes:

$$nv\theta_{ij} = \left(nvl_{ij}, nvm_{ij}, nvr_{ij}\right) = \left(1 - vr_{ij}, 1 - vm_{ij}, 1 - vl_{ij}\right) \tag{13}$$

B. COMPUTE THE WEIGHT THROUGH MEREC APPROACH

Step 3: MEREC approach [68] is illustrated the weight under HTFNs

(1) Construct the normalized hesitant triangular fuzzy number score matrix (NHTFNSM):

$$NHTFNSM\left(nv\theta_{ij}\right) = \frac{\min\limits_{i}\left(SF\left(nv\theta_{ij}\right) + 1\right)}{\left(SF\left(nv\theta_{ij}\right) + 1\right)} \tag{14}$$

(2) Administrate the overall performance $NHTFNSM\ (nv\theta_i)$.

$$NHTFNSM\ (nv\theta_i) = \ln\left(1 + \left(\frac{1}{n}\sum_{j=1}^{n}\left|\ln NHTFNSM\left(nv\theta_{ij}\right)\right|\right)\right),\ i = 1, 2, \cdots, m. \tag{15}$$

(3) Administrate the performance of VA_i through removing the attribute.

$$\begin{aligned} &NHTFNSM\left(nv\theta_{i(j)}\right) \\ &= \ln\left(1 + \left(\frac{1}{n}\sum_{k=1,k\neq j}^{n}\left|\ln NHTFNSM\ (nv\theta_{ik})\right|\right)\right) \\ &i = 1, 2, \cdots, m; k = 1, 2, \cdots, n, \end{aligned} \tag{16}$$

(4) Administrate the summation of absolute deviations:

$$HTFSD_j = \sum_{i=1}^{m}\left|NHTFNSM\left(nv\theta_{i(j)}\right) - NHTFNSM\ (nv\theta_i)\right|, \quad k = 1, 2, \cdots n \tag{17}$$

(5) Administrate the weight:

$$vw_j = \frac{HTFSD_j}{\sum_{j=1}^{n} HTFSD_j}, \quad j = 1, 2, \cdots n. \tag{18}$$

C. HTFN-ExpTODIM-IGRA APPROACH FOR MADM

The HTFN-ExpTODIM-IGRA approach is constructed for MADM.

Step 4: Construct relative weight of HG_j as:

$$rvw_j = vw_j \Big/ \max_j vw_j, \tag{19}$$

Step 5: Construct the overall HTFN dominance degree values (HTFNDDV).

$$HTFHD\left(v\theta_1, v\theta_2\right) = \frac{1}{\#v\theta}\sum_{k=1}^{\#v\theta}\frac{\left(\begin{array}{l}\left|ol_1\left(\sigma\left(k\right)\right) - ol_2\left(\sigma\left(k\right)\right)\right| + \left|om_1\left(\sigma\left(k\right)\right) - om_2\left(\sigma\left(k\right)\right)\right| \\ + \left|or_1\left(\sigma\left(k\right)\right) - or_2\left(\sigma\left(k\right)\right)\right|\end{array}\right)}{3} \tag{8}$$

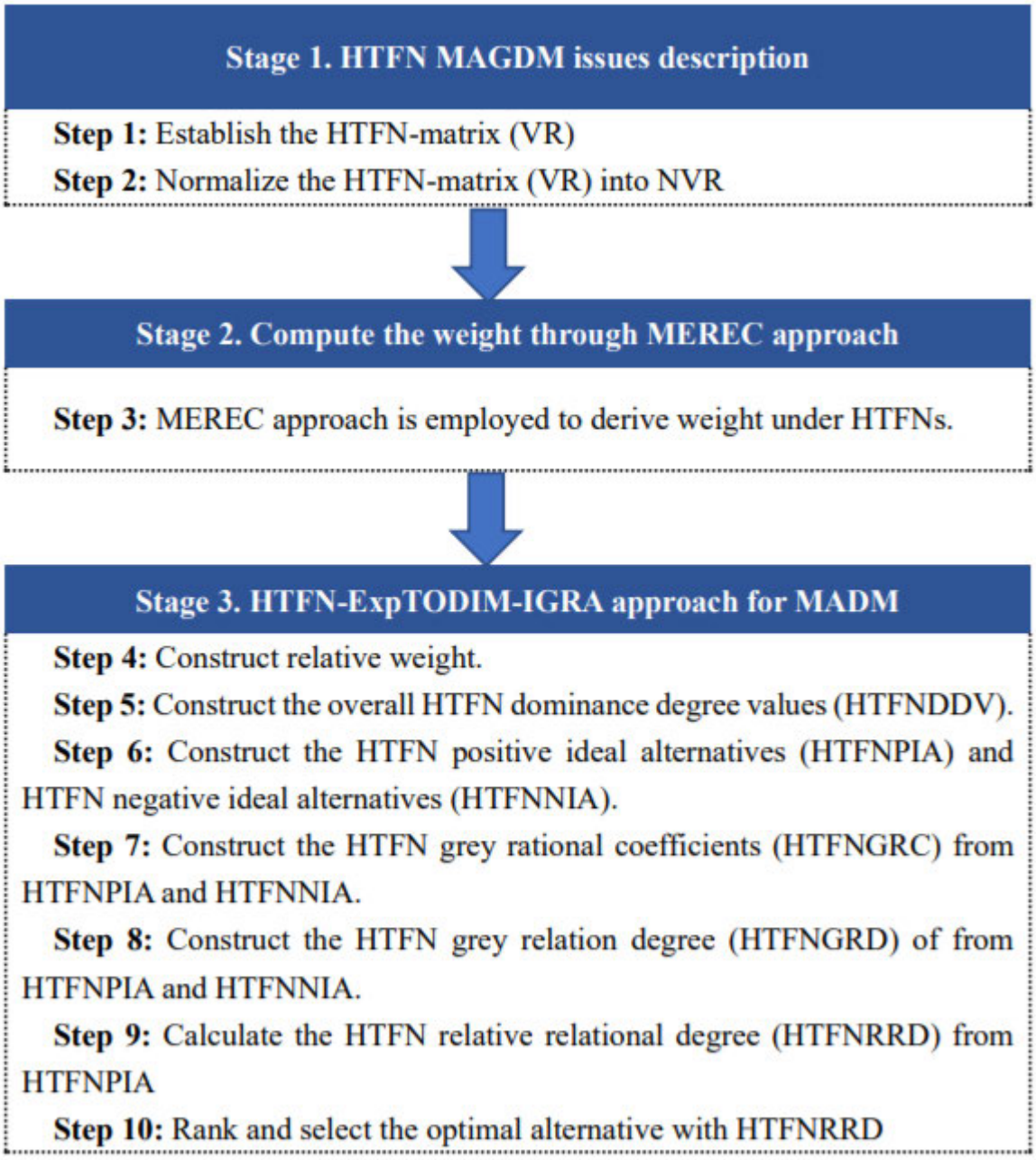


FIGURE 1. Framework of HTFN-ExpTODIM-IGRA approach for MADM with MEREC.

(1) The HTFNDDV of VA_i over VA_t for VG_j is administrated. (20), as shown at the bottom of the next page, where π is constructed from [69] and $\pi \in [1, 5]$ [59].

(2) The $HTFNDDV_j(VA_i)$ for VG_j is constructed: (21), as shown at the bottom of the next page,

(3) Produce the overall HTFNDDV of VA_i over other alternatives under VG_j:

$$HTFNDDV_j(VA_i) = \sum_{t=1}^{m} HTFNDDV_j(VA_i, VA_t) \quad (22)$$

(4) The overall HTFNDDV is constructed: (23), as shown at the bottom of the next page.

Step 6: Construct the HTFN positive ideal alternatives (HTFNPIA) and HTFN negative ideal alternatives (HTFNNIA):

$$HTFNPIA = (HTFNPIA_1, HTFNPIA_1, \cdots, HTFNPIA_n) \quad (24)$$

$$HTFNNIA = (HTFNNIA_1, HTFNNIA_1, \cdots, HTFNNIA_n) \quad (25)$$

$$HTFNPIA_j = \max_{j=1}^{n} \left(HTFNDDV_{ij}\right),$$
$$HTFNNIA_j = \min_{j=1}^{n} \left(HTFNDDV_{ij}\right) \tag{26}$$

Step 7: Construct the HTFNGRC (HTFN grey rational coefficients) from HTFNPIA and HTFNNIA: (27) and (28), as shown at the bottom of the next page, where $\rho = 0.5$.

Step 8: Construct the HTFNGRD (HTFN grey relation degree) of from HTFNPIA and HTFNNIA.

$$HTFNGRD\left(\xi_i^+\right) = \sum_{j=1}^{n} vw_j HTFNGRC\left(\xi_{ij}^+\right) \tag{29}$$

$$HTFNGRD\left(\xi_i^-\right) = \sum_{j=1}^{n} vw_j HTFNGRC\left(\xi_{ij}^-\right) \tag{30}$$

Step 9: Calculate the HTFN relative relational degree (HTFNRRD) from HTFNPIA.

$$HTFNRRD\left(\xi_i\right) = \frac{HTFNGRD\left(\xi_i^+\right)}{HTFNGRD\left(\xi_i^-\right) + HTFNGRD\left(\xi_i^+\right)} = \frac{\sum_{j=1}^{n} vw_j HTFNGRC\left(\xi_{ij}^+\right)}{\sum_{j=1}^{n} vw_j HTFNGRC\left(\xi_{ij}^+\right) + \sum_{j=1}^{n} vw_j HTFNGRC\left(\xi_{ij}^-\right)} \tag{31}$$

Step 10: Rank and select the optimal alternative with HTFNRRD and the optimal alternative has maximum HTFNRRD.

IV. NUMERICAL EXAMPLE AND COMPARATIVE ANALYSIS

A. NUMERICAL EXAMPLE

In the process of optimizing the comprehensive quality evaluation system for students in sports colleges, the primary task is to implement comprehensive evaluation. This means not only evaluating students' academic performance, but also strengthening the evaluation of their moral qualities, practical abilities, and other aspects, in order to comprehensively understand and improve their comprehensive qualities. Firstly, it is necessary to recognize that comprehensive quality does not only refer to professional performance, but also includes students' moral quality, practical ability, innovative spirit, and other aspects. Therefore, the evaluation system

$$HTFNDDV_j\left(VA_i, VA_t\right) = \begin{cases} \dfrac{rvw_j \times \left(1 - 10^{-\rho HTFED\left(nv\theta_{ij}, nv\theta_{tj}\right)}\right)}{\sum_{j=1}^{n} rvw_j} & if\ SF\left(nv\theta_{ij}\right) > SF\left(nv\theta_{tj}\right) \\ 0 & if\ SF\left(nv\theta_{ij}\right) = SF\left(nv\theta_{tj}\right) \\ -\dfrac{1}{\pi}\dfrac{\sum_{j=1}^{n} rvw_j \times \left(1 - 10^{-\rho HTFED\left(nv\theta_{ij}, nv\theta_{tj}\right)}\right)}{rvw_j} & if\ SF\left(nv\theta_{ij}\right) < SF\left(nv\theta_{tj}\right) \end{cases} \tag{20}$$

$$HTFNDDV_j\left(HA_i\right) = \left[HTFNDD_j\left(HA_i, HA_t\right)\right]_{m\times m} = \begin{array}{c} \\ VA_1 \\ VA_2 \\ \vdots \\ VA_m \end{array} \begin{array}{c} \begin{array}{cccc} VA_1 & VA_2 & \cdots & VA_m \end{array} \\ \begin{bmatrix} 0 & HTFNDDV_j\left(VA_1, VA_2\right) & \cdots & HTFNDDV_j\left(VA_1, VA_m\right) \\ HTFNDDV_j\left(VA_2, VA_1\right) & 0 & \cdots & HTFNDDV_j\left(VA_2, VA_m\right) \\ \vdots & \vdots & \cdots & \vdots \\ HTFNDDV_j\left(VA_m, VA_1\right) & HTFNDDV_j\left(VA_m, VA_2\right) & \cdots & 0 \end{bmatrix} \end{array} \tag{21}$$

$$HTFNDDV = \left(HTFNDDV_{ij}\right)_{m\times n} = \begin{bmatrix} & VG_1 & VG_2 & \cdots & VG_n \\ VA_1 & \sum_{t=1}^{m} HTFNDDV_1\left(VA_1, VA_t\right) & \sum_{t=1}^{m} HTFNDDV_2\left(VA_1, VA_t\right) & \cdots & \sum_{t=1}^{m} HTFNDDV_n\left(VA_1, VA_t\right) \\ VA_2 & \sum_{t=1}^{m} HTFNDDV_1\left(VA_2, VA_t\right) & \sum_{t=1}^{m} HTFNDDV_2\left(VA_2, VA_t\right) & \cdots & \sum_{t=1}^{m} HTFNDDV_n\left(VA_2, VA_t\right) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ VA_m & \sum_{t=1}^{m} HTFNDDV_1\left(VA_m, VA_t\right) & \sum_{t=1}^{m} HTFNDDV_2\left(VA_m, VA_t\right) & \cdots & \sum_{t=1}^{m} HTFNDDV_n\left(VA_m, VA_t\right) \end{bmatrix} \tag{23}$$

TABLE 1. HTFN-matrix $VR = \left[v\theta_{ij}\right]_{5\times4}$.

	VG_1	VG_2
VA_1	{(0.37,0.46,0.53), (0.42,0.58,0.67)}	{(0.53,0.67,0.78)}
VA_2	{(0.23,0.37,0.49)}	{(0.37,0.46,0.53), (0.68,0.76,0.83)}
VA_3	{(0.45,0.61,0.68)}	{(0.32,0.48,0.57)}
VA_4	{(0.34,0.47,0.63)}	{(0.34,0.48,0.59), (0.36,0.49,0.59)}
VA_5	{(0.39,0.45,0.62), (0.63,0.79,0.91)}	{(0.46,0.59,0.64),(0.54,0.62,0.74), (0.57,0.62,0.73)}

	VG_3	VG_4
VA_1	{(0.53,0.69,0.78)}	{(0.59,0.64,0.72), (0.63,0.74, 0.81)}
VA_2	{(0.74,0.82,0.84)}	{(0.52,0.58,0.64), (0.72,0.76,0.83)}
VA_3	{(0.53,0.61,0.72)}	{(0.34,0.47,0.58), (0.53,0.67,0.72)}
VA_4	{(0.28,0.47,0.53)}	{(0.38,0.49,0.53), (0.62,0.74,0.85)}
VA_5	{(0.61,0.75,0.82), (0.54,0.63,0.72)}	{(0.27,0.36,0.57)}

needs to achieve comprehensive coverage and cannot focus on one aspect while neglecting other aspects. This requires redesigning and adjusting the evaluation system to ensure that it can comprehensively reflect students' various abilities and qualities. Secondly, the evaluation of moral qualities can be carried out by observing and recording students' behavior and performance in daily life and learning. For example, whether they respect their teachers, whether they are honest and trustworthy, and whether they are willing to help others. At the same time, various moral education activities such as themed class meetings and moral lectures can also be carried out to cultivate and improve students' moral qualities. Once again, for the evaluation of practical abilities, it is necessary to provide more practical opportunities and use practical achievements as an important basis for evaluation. For example, various practical activities such as internships, training, project research, etc. can be carried out, and students' practical abilities can be evaluated based on their performance in these activities. At the same time, industry experts can also be introduced to participate in the evaluation, improving the effectiveness and fairness of the evaluation. Establishing a scientific and reasonable comprehensive quality evaluation system for students in sports colleges and universities needs to follow the following design principles: firstly, the principle of ''comprehensiveness''. The evaluation system should cover students' knowledge, skills, attitudes, values, and other aspects to achieve comprehensive evaluation of students' development. This means that we should not only focus on students' academic performance, but also on their moral character, innovative spirit, practical ability, teamwork and communication skills. The second is the principle of ''fairness''. The evaluation system should ensure that every student is evaluated in a fair and just environment, avoiding subjective bias or unfair evaluation methods from affecting the evaluation results. The evaluation criteria should be open and transparent, and all evaluation processes should have clear regulations and guidelines. The third principle is ''openness''. The evaluation system should adapt to changes in the educational environment and social needs, with a certain degree of flexibility and adaptability. The evaluation system should not be fixed, but should be regularly revised and updated to meet the development needs of society and the

$$HTFNGRC\left(\xi_{ij}^{+}\right) = \frac{\min\limits_{1\le i\le m}\left|HTFNDDV_{ij} - HTFNPIA_j\right| + \rho\max\limits_{1\le i\le m}\left|HTFNDDV_{ij} - HTFNPIA_j\right|}{\left|HTFNDDV_{ij} - HTFNPIA_j\right| + \rho\max\limits_{1\le i\le m}\left|HTFNDDV_{ij} - HTFNPIA_j\right|} \tag{27}$$

$$HTFNGRC\left(\xi_{ij}^{-}\right) = \frac{\min\limits_{1\le i\le m}\left|HTFNDDV_{ij} - HTFNNIA_j\right| + \rho\max\limits_{1\le i\le m}\left|HTFNDDV_{ij} - HTFNNIA_j\right|}{\left|HTFNDDV_{ij} - HTFNNIA_j\right| + \rho\max\limits_{1\le i\le m}\left|HTFNDDV_{ij} - HTFNNIA_j\right|} \tag{28}$$

TABLE 2. The $HN = \left[nh\theta_{ij}\right]_{5\times4}$.

	VG_1
VA_1	{(0.37,0.46,0.53), (0.37,0.46,0.53), (0.42,0.58,0.67)}
VA_2	{(0.23,0.37,0.49), (0.23,0.37,0.49), (0.23,0.37,0.49)}
VA_3	{(0.45,0.61,0.68), (0.45,0.61,0.68), (0.45,0.61,0.68)}
VA_4	{(0.34,0.47,0.63), (0.34,0.47,0.63), (0.34,0.47,0.63)}
VA_5	{(0.39,0.45,0.62), (0.39,0.45,0.62), (0.63,0.79,0.91)}

	VG_2
VA_1	{(0.53,0.67,0.78), (0.53,0.67,0.78), (0.53,0.67,0.78)}
VA_2	{(0.37,0.46,0.53), (0.37,0.46,0.53), (0.68,0.76,0.83)}
VA_3	{(0.32,0.48,0.57), (0.32,0.48,0.57), (0.32,0.48,0.57)}
VA_4	{(0.34,0.48,0.59), (0.34,0.48,0.59), (0.36,0.49,0.59)}
VA_5	{(0.46,0.59,0.64), (0.54,0.62,0.74), (0.57,0.62,0.73)}
	VG_3
VA_1	{(0.53,0.69,0.78), (0.53,0.69,0.78), (0.53,0.69,0.78)}
VA_2	{(0.74,0.82,0.84), (0.74,0.82,0.84), (0.74,0.82,0.84)}
VA_3	{(0.53,0.61,0.72), (0.53,0.61,0.72), (0.53,0.61,0.72)}
VA_4	{(0.28,0.47,0.53), (0.28,0.47,0.53), (0.28,0.47,0.53)}
VA_5	{(0.54,0.63,0.72), (0.54,0.63,0.72), (0.61,0.75,0.82), }

	VG_4
VA_1	{(0.59,0.64,0.72), (0.59,0.64,0.72), (0.63,0.74, 0.81)}
VA_2	{(0.52,0.58,0.64), (0.52,0.58,0.64), (0.72,0.76,0.83)}
VA_3	{(0.34,0.47,0.58), (0.34,0.47,0.58), (0.53,0.67,0.72)}
VA_4	{(0.38,0.49,0.53), (0.38,0.49,0.53), (0.62,0.74,0.85)}
VA_5	{(0.27,0.36,0.57), (0.27,0.36,0.57), (0.27,0.36,0.57)}

industry. The fourth is the principle of ''dynamism''. The evaluation system should reflect students' learning process and growth changes, rather than just focusing on their final grades. By continuously tracking and evaluating students, it is possible to have a more accurate understanding and promote their comprehensive development. The fifth principle is practicality. The design of the evaluation system should be easy to implement, and the evaluation criteria should be clear and operable. At the same time, the evaluation results should provide effective feedback for teaching decisions and students' learning. The comprehensive quality evaluation of physical education institutions is a classical MADM issue. Five physical education colleges VA_i $(i = 1, 2, 3, 4, 5)$ are chosen with four attributes: ① GG_1 is knowledge and skills.

TABLE 3. The $HTFNDDV = \left(HTFNDDV_{ij}\right)_{5\times4}$.

	VG_1	VG_2	VG_3	VG_4
VA_1	0.4801	-0.1832	-0.6361	-0.4074
VA_2	-1.0994	0.5649	0.8350	-0.1233
VA_3	0.0448	1.0157	-0.1850	0.1959
VA_4	0.4283	0.4999	-0.1165	0.9640
VA_5	0.8391	-0.6717	-0.5550	-0.7883

TABLE 4. The HTFNPIA and HTFNNIA.

	VG_1	VG_2	VG_3	VG_4
HTFNPIA	0.8391	1.0157	0.8350	0.9640
HTFNNIA	-1.0994	-0.6717	-0.6361	-0.7883

TABLE 5. The $HTFNGRC\left(\xi_{ij}^{+}\right)$.

	VG_1	VG_2	VG_3	VG_4
VA_1	0.7297	0.4131	0.3333	0.3898
VA_2	0.3333	0.6518	1.0000	0.4462
VA_3	0.5496	1.0000	0.4190	0.5329
VA_4	0.7023	0.6206	0.4360	1.0000
VA_5	1.0000	0.3333	0.3461	0.3333

TABLE 6. The $HTFNGRC\left(\xi_{ij}^{-}\right)$.

	VG_1	VG_2	VG_3	VG_4
VA_1	0.3803	0.6333	1.0000	0.6970
VA_2	1.0000	0.4056	0.3333	0.5685
VA_3	0.4586	0.3333	0.6199	0.4710
VA_4	0.3882	0.4186	0.5860	0.3333
VA_5	0.3333	1.0000	0.9007	1.0000

Knowledge and skills are foundation of comprehensive qualities, mainly referring to the professional knowledge and skills acquired by students during their studies in physical education institutions. It not only includes mastery of theoretical knowledge, but also proficiency in skill operations and the ability to independently solve problems. ② GG_2 is moral quality. Moral quality is an important indicator for evaluating the comprehensive quality of vocational college students, mainly reflected in adhering to social ethics, professional ethics, and school rules and regulations, respecting

TABLE 7. The *HTFNGRD* $\left(\xi_i^+\right)$, *HTFNGRD* $\left(\xi_i^-\right)$.

	$HTFNGRD\left(\xi_i^+\right)$	$HTFNGRD\left(\xi_i^-\right)$
VA_1	0.4845	0.6400
VA_2	0.5927	0.5877
VA_3	0.6877	0.4447
VA_4	0.6710	0.4274
VA_5	0.5254	0.7917

TABLE 8. The *HTFNRRD* (ξ_i).

	$HTFNRRD\left(\xi_i\right)$	Order
VA_1	0.4309	4
VA_2	0.5021	3
VA_3	0.6073	2
VA_4	0.6109	1
VA_5	0.3989	5

TABLE 9. Order for different approaches.

	Order
HTFWA approach[66]	$VA_4 > VA_3 > VA_2 > VA_1 > VA_5$
HTFWG approach[66]	$VA_4 > VA_3 > VA_1 > VA_2 > VA_5$
HTFEWA approach [72]	$VA_4 > VA_3 > VA_2 > VA_1 > VA_5$
HTFEWG approach [72]	$VA_4 > VA_3 > VA_1 > VA_2 > VA_5$
HTFECA approach [73]	$VA_4 > VA_3 > VA_2 > VA_1 > VA_5$
HTFCOG approach [74]	$VA_4 > VA_3 > VA_1 > VA_2 > VA_5$
HTFWBM approach [75]	$VA_4 > VA_3 > VA_2 > VA_1 > VA_5$
HTFGWBM approach [76]	$VA_4 > VA_3 > VA_2 > VA_1 > VA_5$
HTF-TODIM approach [77]	$VA_4 > VA_3 > VA_2 > VA_1 > VA_5$
HTFN-ExpTODIM-IGRA approach	$VA_4 > VA_3 > VA_2 > VA_1 > VA_5$

others, and having a good sense of social responsibility and professional ethics. ③ GG_3 is spirit of innovation. The spirit of innovation is an important requirement for students in sports colleges and universities in today's society. This is mainly manifested as students having the ability to think independently, solve problems, and innovate new knowledge. ④ GG_4 is practical ability. Practical ability is one of core qualities of students in sports colleges, emphasizing the ability of students to apply their knowledge to practice and solve practical problems. Five possible physical education colleges are evaluated through HTFNs.

The HTFN-ExpTODIM-IGRA approach is put forward the comprehensive quality evaluation of physical education institutions.

Step 1: Illustrate the HTFN-matrix $VR = \left[v\theta_{ij}\right]_{5\times4}$ (Table 1).

TABLE 10. WS coefficient information for different approaches.

	HTFN-ExpTODIM-IGRA approach	HTFNWA approach	HTFNWG approach	HTFNEWA approach
VA_1	4	4	4	4
VA_2	3	3	3	3
VA_3	2	2	1	2
VA_4	1	1	2	1
VA_5	5	5	5	5
Coefficients	WS	1.0000	0.7917	1.0000

	HTFN-ExpTODIM-IGRA approach	HTFNEWG approach	HTFNECA approach	HTFNCOG approach
VA_1	4	4	4	4
VA_2	3	3	3	3
VA_3	2	1	2	1
VA_4	1	2	1	2
VA_5	5	5	5	5
Coefficients	WS	0.7917	1.0000	0.7917

	HTFN-ExpTODIM-IGRA approach	HTFNWBM approach	HTFNGWBM approach	HTFN-TODIM approach
VA_1	4	4	4	4
VA_2	3	3	3	3
VA_3	2	2	2	2
VA_4	1	1	1	1
VA_5	5	5	5	5
Coefficients	WS	1.0000	1.0000	1.0000

Step 2: Normalize the $HR = \left[h\theta_{ij}\right]_{5\times4}$ into $HN = \left[nh\theta_{ij}\right]_{5\times4}$ (See Table 2).

Step 3: Compute the weight:

$$vw_1 = 0.2846, \quad vw_2 = 0.3667,$$
$$vw_3 = 0.1865, \quad vw_4 = 0.1622.$$

Step 4: Produce the relative weight: $rvw = \{0.7761, 1.0000, 0.5086, 0.4423\}$

Step 5: Administrate the $HTFNDDV = \left(HTFNDDV_{ij}\right)_{5\times4}$ (See Table 3):

Step 6: Construct the HTFNPIA and HTFNNIA (See table 4).

Step 7: Illustrate the $HTFNGRC\left(\xi_{ij}^{+}\right)$ and $HTFNGRC\left(\xi_{ij}^{-}\right)$ (Table 5-6).

Step 8: Construct the $HTFNGRD\left(\xi_{i}^{+}\right)$ and $HTFNGRD\left(\xi_{i}^{-}\right)$ (Table 7).

Step 9: Calculate the $HTFNRRD\left(\xi_i\right)$ (See table 7).

Step 10: In light with HTFNRRD, the order is: $VA_4 > VA_3 > VA_2 > VA_1 > VA_5$ and the optimal physical education college is VA_4.

B. COMPARATIVE ANALYSIS

The HTFN-ExpTODIM-IGRA approach is compared with HTFWA approach [65] and HTFWG approach

[65], HTFEWA approach [70], HTFEWG approach [70], HTFECA approach [71], HTFCOG approach [72], HTFWBM approach [73], HTFGWBM approach [74] and HTFN-TODIM approach [75]. The final comparative results are depicted in Table 8.

Furthermore, the order similarity between HTFNWA approach [65], HTFNWG approach [65], HTFNEWA approach [70], HTFNEWG approach [70], HTFN-ECA approach [71], HTFNCOG approach [72], HTFN-WBM approach [73], HTFNGWBM approach [74] and HTFN-TODIM approach [75] and HTFN-ExpTODIM-IGRA approach was analyzed through WS coefficients [76], the final results are managed in Table 9.

From the analysis, it is evident that the ranking order of the physical education colleges produced by the different approaches may vary slightly. However, importantly, all the approaches identify the same optimal and worst-performing physical education colleges. This consistency verifies that the proposed HTFN-ExpTODIM-IGRA approach is effective for the comprehensive quality evaluation. The major advantage of HTFN-ExpTODIM-IGRA approach can be summarized: HTFN-ExpTODIM-IGRA approach not only captures the decision-makers' psychological behavior, but also accounts for shape similarity between the HTFPIS and HTFNIS during the comprehensive quality evaluation of physical education institutions. This holistic consideration of both subjective and objective factors enhances the robustness of the decision-making process.

V. CONCLUSION

With the development of society and economic transformation, the comprehensive quality of students in sports colleges is receiving increasing attention. At present, there are problems in the comprehensive quality evaluation system of students in sports colleges, such as excessive emphasis on academic performance, fixed evaluation system, lack of student participation, and unclear evaluation standards. These problems have caused certain obstacles to the comprehensive development of vocational college students. Therefore, it is necessary to conduct comprehensive evaluation of students' moral qualities, practical abilities, and other aspects, regularly revise the evaluation system, introduce student self-evaluation and mutual evaluation mechanisms, and develop clear, specific, and quantifiable evaluation standards. Research the constituent elements of the comprehensive quality of physical education institutions, propose scientific and reasonable evaluation system implementation steps, and propose optimization suggestions for the current problems, in order to promote the comprehensive development of students and the reform of physical education. The comprehensive quality evaluation of physical education institutions involves MADM. ExpTODIM and IGRA approach have been illustrated for MADM problems. HTFSs have been employed as useful approach for characterizing uncertain data during the comprehensive quality evaluation of physical education institutions. In this research, the HTFN-ExpTODIM-IGRA approach is illustrated to solve the MADM problem under the framework of HTFSs. The major contributions of this work are outlined:(1) MEREC approach is put forward weight with HTFSs; (2) HTFN-ExpTODIM-IGRA approach is developed to address the MADM; (3) numerical example for comprehensive quality evaluation of physical education institutions is provided to validate the HTFN-ExpTODIM-IGRA approach. The paper demonstrates the application of advanced MADM techniques, such as ExpTODIM and IGRA, in combination with the HTFS model, to tackle the complex problem of comprehensive quality evaluation in the domain of physical education institutions. The developed methodologies provide a structured framework for handling uncertainty and MADM in this context.

While the proposed HTFN-ExpTODIM-IGRA approach demonstrates its applicability in the comprehensive quality evaluation of physical education institutions, there are some potential limitations that could be addressed in future research: (1) Consensus modeling: Incorporating consensus building mechanisms into the comprehensive quality evaluation under the HTFS framework could be a worthwhile research direction. This would help capture the collective opinions and preferences of stakeholders involved in the evaluation process. (2) Integration of regret theory: Combining the proposed methods with regret theory could provide additional insights into the decision-making process for comprehensive quality evaluation. Regret-based approaches may offer a more robust handling of uncertainties and risk preferences. (3) Exploring other fuzzy extensions: In subsequent research, the integration of TODIM approach with other decision-making methods employing different fuzzy extensions, could be investigated. This could further enhance the flexibility and applicability of the comprehensive quality evaluation framework for physical education institutions.

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RUI LU was born in Guizhou, China. He received the Bachelor of Education degree from Chengdu University of Physical Education, in 2004. He is currently a Lecturer with the School of Physical Education, Guizhou University of Finance and Economics. He has been engaged in college physical education and sports competition, since 2004. His main research interest includes sports training.



GUOXIAN GAO was born in Henan, China. He received the master's degree in physical education and training and the Ph.D. degree in education from Beijing Sport University, in September 2007 and in June 2018, respectively. He has published six theoretical results in journals, such as *Sports Science*, *Sports Journal*, and *Journal of Shenyang Institute of Physical Education*, and presided more than one provincial and ministerial project. His research interest includes basketball theory and practice.



YUHONG XIANG received the Doctor of Education degree in sports humanities and sociology from Beijing Sport University. He is an Associate Professor and a Master's Tutor with the Physical Education Department, Guizhou University of Finance and Economics, and a Researcher with the Key Laboratory of Big Data, Guizhou University of Finance and Economics, and the Special Expert of Guizhou Sports Tourism Research Institute. His main research interests include related research on sports humanities and sociology and folk sports.